

(a) Model unit and sequence windows

Model unit
fixed oligomer slots
default 8

1. ABCDABDDC
2. CABDABADC
3. ABCDABDDC
4. DCABACDBA
5. BACDABDDC
6. ABDDABADC
7. DCABACDBA
8. CABDABADC

Illustrative oligomer sequence
(not the full model length)
ABDCABDDCABD

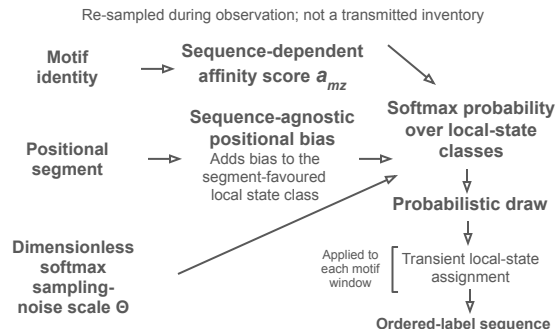
ABDCA $u = 1$
BDCAB $u = 2$
DCABB $u = 3$
CABBD $u = 4$
Motif-window start index u

4 equal positional bins over motif-window start positions

S1 S2 S3 S4
 $u = 1-14$ $u = 15-28$ $u = 29-42$ $u = 43-56$

Default favoured local-state order by segment: 2, 0, 3, 1
Sequence identities are distinct from transient local-state labels

(b) Transient local-state assignment



(c) Model fitness

Ordered-label sequence 0-1-3-2-0

Adjacent pairs & classification
(0,1) (1,3) (3,2) (2,0)
P R P N

Adjacency-category matrix
(first label z , next label z')

		z'			
		0	1	2	3
z	0	R	P	N	P
	1	P	R	P	R
	2	N	P	R	P
	3	P	R	P	R

P = fitness-promoting
R = fitness-reducing
N = fitness-neutral

Mean fitness-promoting
adjacency count per
oligomer $n_{i,t}^+$

Mean fitness-reducing
adjacency count per
oligomer $n_{i,t}^-$

Model fitness

Phenomenological score
not protocell
growth chemistry

(d) Population update

Select one source model
unit according to fitness

Selected source model unit

with probability p : sample with replacement
a parent-derived sequence identity from the
selected source unit's empirical distribution

with probability $1 - p$: sample a
sequence identity from the idealized
uniform sequence reservoir

parental-composition coupling p
(compositional-transmission fidelity)

point mutation

New model unit

The same source sequence
identity may populate more
than one new-unit slot

Population update = fitness-weighted source selection + independent slot-level
compositional resampling + point mutation